Trigonometry

ANSWER KEY

Section 1: Unit Circle & Special Angles

1 What is the exact value of $\sin(150^\circ)$?

a. $\frac{1}{2}$ ✓

b. $-\frac{1}{2}$

c. $\frac{\sqrt{3}}{2}$

d. $-\frac{\sqrt{3}}{2}$

Reference angle in Quadrant II is $180^\circ - 150^\circ = 30^\circ$. Sine is positive in QII: $\sin(150^\circ) = \sin(30^\circ) = \frac{1}{2}$.

2 What is the exact value of $\cos(240^\circ)$?

a. $-\frac{1}{2}$ ✓

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. $\frac{\sqrt{3}}{2}$

Reference angle in Quadrant III is $240^\circ - 180^\circ = 60^\circ$. Cosine is negative in QIII: $\cos(240^\circ) = -\cos(60^\circ) = -\frac{1}{2}$.

3 Convert $\frac{5\pi}{6}$ radians to degrees.

a. 150° ✓

b. 120°

c. 135°

d. 210°

Multiply by $\frac{180^\circ}{\pi}$: $\frac{5 \times 180^\circ}{6} = 5 \times 30^\circ = 150^\circ$.

Section 2: Pythagorean Identities & Simplifications

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4 Simplify the trigonometric expression: $\frac{\sin^2\theta}{1-\cos^2\theta}$.

a. 1 ✓

b. $\tan^2\theta$

c. $\cot^2\theta$

d. $\sin\theta$

From the fundamental identity $\sin^2\theta+\cos^2\theta=1$, we have $1-\cos^2\theta=\sin^2\theta$. Thus $\frac{\sin^2\theta}{\sin^2\theta}=1$.

5 If $\cos\theta=\frac{3}{5}$ and θ is an acute angle in Quadrant I, what is $\tan\theta$?

a. $\frac{4}{3}$ ✓

b. $\frac{3}{4}$

c. $\frac{4}{5}$

d. $\frac{5}{3}$

Using the $3-4-5$ right triangle, $\sin\theta=\frac{4}{5}$. Then $\tan\theta=\frac{\sin\theta}{\cos\theta}=\frac{4/5}{3/5}=\frac{4}{3}$.

6 What is the simplified form of $\sec\theta\cot\theta$?

a. $\csc\theta$ ✓

b. $\sin\theta$

c. $\cos\theta$

d. $\tan\theta$

$\sec\theta\cot\theta=\frac{1}{\cos\theta}\times\frac{\cos\theta}{\sin\theta}=\frac{1}{\sin\theta}=\csc\theta$.

Section 3: Double Angle Formulas & Equations

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7 If $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$, what is the value of $\sin(2\theta)$?

a. $\frac{24}{25}$ ✓

b. $\frac{12}{25}$

c. $\frac{7}{25}$

d. $\frac{18}{25}$

Double-angle identity: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

8 What are the solutions to $2\cos\theta - 1 = 0$ on the interval $[0, 2\pi)$?

a. $\frac{\pi}{3}, \frac{5\pi}{3}$ ✓

b. $\frac{\pi}{6}, \frac{11\pi}{6}$

c. $\frac{2\pi}{3}, \frac{4\pi}{3}$

d. $\frac{\pi}{4}, \frac{7\pi}{4}$

$\cos\theta = \frac{1}{2}$. Positive cosine occurs in QI and QIV: $\theta = \frac{\pi}{3}$ and $\theta = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$.

9 Which expression is equivalent to $\cos(2\theta)$?

a. $\cos^2\theta - \sin^2\theta$ ✓

b. $2\cos^2\theta + 1$

c. $1 - 2\cos^2\theta$

d. $\cos^2\theta + \sin^2\theta$

Standard double-angle formula for cosine is $\cos(2\theta) = \cos^2\theta - \sin^2\theta$.

Section 4: Laws of Sines & Cosines

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10 In $\triangle ABC$, $a = 8$, $\sin A = 0.4$, and $\sin B = 0.6$. What is the length of side b?

a. 12 ✓

b. 10

c. 14

d. 16

Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow \frac{8}{0.4} = \frac{b}{0.6} \Rightarrow 20 = \frac{b}{0.6} \Rightarrow b = 12$.

11 In $\triangle ABC$, $b = 5$, $c = 6$, and $\angle A = 60^\circ$. What is the value of a^2?

a. 31 ✓

b. 61

c. 41

d. 25

Law of Cosines:
$a^2 = b^2 + c^2 - 2bc\cos A = 25 + 36 - 2(5)(6)\cos(60^\circ) = 61 - 60(0.5) = 61 - 30 = 31$.

12 What is the area of a triangle with side lengths 6 cm and 8 cm enclosing an angle of 30°?

a. 12 cm^2 ✓

b. 24 cm^2

c. $12\sqrt{3}$ cm^2

d. 16 cm^2

Area $= \frac{1}{2}ab\sin C = \frac{1}{2}(6)(8)\sin(30^\circ) = 24 \times 0.5 = 12$ cm^2.

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